Benjamín Rodríguez Romo

MASTER OF SCIENCES IN BIOENGINEERING (UAI)|BIOENGINEERING AND INFORMATICS ENGINEERING DEGREES (UAI)| May 14th, 2001| Seidenstrasse 14, Dübendorf, Switzerland| (+41) 79 932 37 09 I benjaminrodriguezromo@gmail.com I https://www.linkedin.com/in/benjamín-rodríguez-romo-865b48230/

I am a Bioengineer specialized in systems and data, with experience in experimental work and data science analysis. I have skills in experimental and computational techniques for manufacturing, biomechanical testing and analysis of hard and soft tissues, and developing artificial intelligence models and pipelines for process improvement. I stand out for my ability to adapt, analyze, and solve problems, with a focus on process optimization and management using digital tools and technology.

PROFESSIONAL EXPERIENCE - RESEARCH

- **ADOLFO IBAÑEZ UNIVERSITY, Msc in Bioengineering (expected)** **2023-2026**
- **ADOLFO IBAÑEZ UNIVERSITY, Professional degree in Bioengineering (expected)** **2019–2026**
- **ADOLFO IBAÑEZ UNIVERSITY, Professional degree in Informatics Engineering (expected)** **2019–2026**

PROFESSIONAL EXPERIENCE - RESEARCH

EMPA, SWISS FEDERAL LABORATORIES FOR MATERIALS SCIENCE AND TECHNOLOGY. DÜBENDORF, SWITZERLAND
Febraury 2026 – Febraury 2027
Intern
- Statistical and Machine Learning analysis.
- Finite element modeling (FEM) using Abaqus.
- Python Programming.
- Sample preparation.

RUSH UNIVERSITY MEDICAL CENTER (BIOMECHANICS DEPARTMENT). CHICAGO, IL, USA
June 2025 – July 2025
Exchange Research Student
- Use of Statistical Shape Models.
- 3D Model alignment and analysis**.**
- Python Programming.

BIOENGINEERING CENTER, B3MAT RESEARCH GROUP, UAI. VIÑA DEL MAR, CHILE
March 2022 - December 2025
Reasearch Assistant (B3MAT)
- Mechanical testing and analysis of mechanical properties in biomaterials.
- 3D manufacturing.
- Imaging processing and analysis from microCT, MRI and X-Ray.
- Support for R&D projects.
- Monitoring and maintenance of laboratory equipment.
- Lab Manager of Biomechanical/Biomaterials Lab.

- Supervision of student’s directed research.

PROFESSIONAL EXPERIENCE - TEACHING

ADOLFO IBÁÑEZ UNIVERSITY August 2022 - December2024

Teaching assitant of the Engineering and Sciences Faculty

- *Bioinformatics and Data Science (2024-2):*
  Assited principal instructor grading tests and projects. This is a mid-level undergradute course.
- *Fundamentals of Data Science (2024-2):*
  Assisted principal instructor with solving problems class and developing and grading tests and homework assignments. This is a beginning level undergraduate course.
- *Experimental Techniques in Biomaterials (Postgraduate) (2024-1):*
  Assisted principal instructor with preparation of samples and laboratories. This is a masters-level course.
- *Biomechanics (2024-1, 2023-1):*
  Assisted principal instructor with preparation of samples and laboratories, and grading tests and homework. This is a mid-level undergraduate course.
- *Biomaterials (2023-2):*
  Assisted principal instructor with preparation of samples and laboratories, and grading tests and homework. This is a mid-level undergraduate course.
- *Bioprocesses (2023-2, 2024-2):*
  Assisted principal instructor with solving problems class and developing and grading tests and homework assignments. This is a beginning level undergraduate course.
- *Introduction to Engineering (2022-2, 2023-1):*
  Assisted principal instructor with grading tests, homework assingments and project presentations. This is a begging-level undergraduate course.
- *Innovation and Entrepreneurship (2022-2):*
  Assisted principal instructor with grading tests, homework assingments and project presentations. This is a begging-level undergraduate course.

Skills - Manufacturing and Laboratory

- Computed Tomography Image and Magnetic Resonance segmentation using 3DSlicer and ImageJ.
- 3D Model reconstruction, visualization, analysis and manipulation with Python using VTK, Open3D, Scipy and Skimage.
- 3D Design using Fusion and Inventor.
- 3D printing with FDM.
- Mechanical testing follows by ASTM, for polymers materials as PLA and ABS, compression, bending and tensile.
- Manufacture of polymer filaments using a conventional extruder.
- Synthesis of materials such as hydroxyapatite and natural calcium carbonate.
- Microscopy image capture and analysis.
- Experimental measurements, porosity, pore size and density.
- Conventional Laser Cutter, MDF and Acrylic.
- Training Artificial Inteligence models for image and biomedical tasks with Python using PyTorch.

Skills - Computer

- Python (Advance).
- 3DSlicer (Advance).
- LaTeX (Advance).
- R (Mid).
- SQL (Mid).
- ImageJ (Mid).

- CellProfiler (Mid).
- Excel (Mid).
- Abaqus (Mid)
- C (Basic).
- Microsoft Azure (Basic).
- Fusion (Basic).

Scolarships and awards

- Academic Honor Scholarship Adolfo Ibáñez University. 2019 - 2023
  For outstanding students with GPA above 6.0 of 7.0.
- PSU Score Scholarship Adolfo Ibáñez University. 2018
  Equivalent to SAT exam. Outstanding Score 760 of 850.
- Third-best graduate, class of 2018 Andres Bello-Pampa HighSchool.
  2018
- Academic Achievement Award Andres Bello-Pampa HighSchool. 2018
- Outstanding athlete Andres Bello-Pampa HighSchool. 2018

Skills - Languages

- Inglés (B2-Mid)
- Spanish (Native speaker)

Skills - Extraprogramatic and Student Activities

- COIL experience for Biomimetics class with Bucknell University. 2025
- Member of Student Organization recognized by the University (Biohuella). 2022-2023
- Basketball Team of Adolfo Ibáñez University. 2019-2022
- Photoshop Workshop. 2020
- Laser cutting machine Workshop. 2019
- Capitan of the youth Basketball selection Team of La Serena City. . 2015-2017

Publications, Conference and Proceedings

- IEEE 15th International Conference on Pattern Recognition Systems (ICPRS 2025). Proceedings and oral presentation. CAMAlyzer: A 3DSlicer extension for AI segmentation and generation of proximal femur 3D models. 1st author.
- Orthopedic Research Society 2025 Annual Meeting (ORS). Conference poster. *Pipeline for MRI automatic segmentation and generation of a 3D model of the femoral head using Machine Learning*. *1st author.*
- Orthopedic Research Society 2026 Annual Meeting (ORS). Conference Poster*. A Framework for Quantifying Tibial Cartilage Deformation After Activity In Vivo via MRI. 2nd author.*
- IEEE Journal of Biomedical and Health Informatics. Journal publication. *CAMalyzer: An Automated MRI-Based 3D Slicer Framework for Femoral Morphology Analysis in Femoroacetabular Impingement Syndrome. First author (Under review)*
- ACS Biomaterials Science & Engineering. *Microstructural heterogeneity of β-TCP granules regulates early stem cell adhesion and spatial organization, revealed by quantitative morphometric analysis. Third author. (Under review*)
- *In preparation. Journal Publication. Photogrametry and 3D printing for replicating complex biological structures like seashells.*